

BUSINESS ANALYTICS

PREDICTING WARRANTY CLAIM LIKELIHOOD FOR HOME APPLIANCES

Based on Usage and Purchase Patterns

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1. Problem Statement and Objectives

1.1 Approved Business Problem Statement

Home appliance retailers and manufacturers in India commonly price extended warranty plans using a single flat rate applied uniformly across all customers, regardless of appliance type, brand, purchase channel, or actual usage behaviour. This one-size-fits-all approach creates two costly problems:

- Revenue leakage from high-risk segments: Customers with heavy daily usage, budget-tier brands, or prior repair histories are under-charged relative to their true claim probability, leading to elevated warranty payout ratios that erode profit margins.
- Missed revenue from low-risk segments: Genuinely low-risk customers · those with light usage, established brands, and authorized-dealer purchases · perceive the flat premium as overpriced and decline the warranty, causing lost ancillary revenue.

Extended warranties carry gross margins of 50-70% in the consumer appliance industry, making them one of the largest sources of high-margin ancillary revenue for retailers. Mispricing directly hurts both customer acquisition and profitability.

1.2 Business Importance and Context

Addressing this pricing asymmetry is critical across the appliance retail ecosystem:

- For Appliance Retailers & Manufacturers: Enables risk-based pricing tiers that improve warranty plan profitability while increasing uptake among low-risk customers.
- For Warranty/Insurance Underwriters: Provides improved actuarial accuracy for premium calculation and claims reserve estimation.
- For Consumers: Delivers pricing that better reflects individual risk profiles · fair premiums for careful users, appropriate surcharges for high-risk profiles.

1.3 Specific Objectives

1. Identify Key Risk Drivers: Determine and rank the usage, purchase, and demographic factors that most strongly influence warranty claim likelihood across four major appliance categories (Refrigerator, Washing Machine, Air Conditioner, Microwave).
2. Build a Claim-Risk Classification Model: Develop and evaluate Decision Tree and Logistic Regression classifiers predicting whether a customer/appliance profile is low-, medium-, or high-claim-risk.
3. Compare Risk Patterns Across Segments: Analyze claim-risk variations across appliance categories, brands (LG, Samsung, Whirlpool, Godrej, Haier, Voltas), and purchase channels (Authorized Dealer vs. Online Marketplace vs. Unauthorized Reseller).
4. Deliver Actionable Pricing Recommendations: Translate classification outputs and feature importance rankings into practical, tiered warranty pricing recommendations.

2. Data Collection and Dataset Description

2.1 Data Collection Methodology & Pipeline Architecture

In compliance with the case study guidelines, no pre-packaged datasets from Kaggle, UCI, GitHub repositories, or similar sources were used as the primary dataset. Data collection followed a multi-stage pipeline combining primary survey collection with structured augmentation and supplementary validation from public consumer grievance platforms.

Stage 1: Primary Survey · Google Forms Questionnaire

A 15-question structured online questionnaire was designed and deployed via Google Forms to capture warranty-relevant attributes from household appliance owners. The survey targeted owners of four major appliance categories: Refrigerators, Washing Machines, Air Conditioners, and Microwaves.

- Target Population: Household appliance owners across 12+ Indian cities.
- Distribution Channels: Personal networks, WhatsApp appliance-owner groups, residential apartment communities, and college peer groups.
- Geographic Coverage: Coimbatore, Chennai, Hyderabad, Tirupati, Mumbai, Kerala, Gujarat, Salem, Trichy, Theni, Pollachi, Anantapur, and more.
- Collection Period: September 17-20, 2026 (active survey window).
- Response Volume: 72 verified responses with complete profiles.
- Data Retrieval: Responses programmatically downloaded via Google Forms API using custom Python CLI (`google_forms_cli.py`) with OAuth 2.0 authentication.

Stage 2: Supplementary Validation · Consumer Complaint Mining

To validate distributional patterns observed in the primary survey, publicly available consumer complaint and review data was cross-referenced from the following sources:

- ConsumerComplaints.in · India's largest consumer grievance platform. Warranty-related complaint threads for LG, Samsung, Whirlpool, Godrej, and Haier appliances were reviewed to confirm brand-level claim frequency patterns.
- MouthShut.com · Product reviews and after-sales service ratings for home appliances were cross-referenced to validate customer satisfaction score distributions.
- National Consumer Helpline (`consumerhelpline.gov.in`) · Official government portal data on category-wise consumer grievance filings was used to confirm that washing machines and air conditioners attract disproportionately higher service complaints.

Stage 3: Stratified Data Augmentation

The 72 primary survey responses were augmented to 12,000 records using a stratified bootstrapping and synthetic generation pipeline that preserves the original distributional characteristics:

1. Stratified Resampling: Survey distributions across appliance categories, brands, and channels were maintained during bootstrap expansion.
2. Statistical Noise Injection: Continuous variables were perturbed with controlled Gaussian noise to prevent exact duplicate rows.
3. Domain-Consistent Feature Engineering: Nine additional features were computed from the base attributes (`Usage_Intensity_Score`, `Annual_Usage_Hours`, `Claim_Probability`, `Claim_Risk_Level`, `Suggested_Annual_Premium_INR`).

Stage 4: Data Persistence & Quality Assurance

All datasets were version-controlled with Git. Raw survey responses were preserved in their original format (`google_form_responses.csv`). The augmentation pipeline produced both raw and preprocessed variants, and an Excel workbook was generated for stakeholder-friendly review.

2.2 Dataset Scale: Raw vs. Cleaned

Dataset File	Records	Attributes	Description
google_form_responses.csv	72	15	Raw primary survey responses
Warranty_Claim_Raw_12000.csv	12,000	24	Augmented with engineered features
Warranty_Claim_Preprocessed.csv	12,000	24	Cleaned, null-imputed, encoded
Warranty_Claim_Analytics.xlsx	12,000	24	Excel formatted analytics tables

2.3 Key Attributes & Variable Categories (24 Variables)

Category	Variables	Description
Identifiers	Appliance_ID, Purchase_Date	Unique record identifiers
Appliance Profile	Category, Brand, Channel, Region, City, Price, Age	Product & purchase characteristics
Usage Patterns	Household_Size, Daily/Annual_Usage_Hours, Intensit	Behavioral wear indicators
Warranty & Claims	Service_Calls, Repair_History, Claims, Outcomes	Warranty event history
Risk Metrics	Claim_Probability, Risk_Level, Annual_Premium	Engineered risk segmentation
Satisfaction	Customer_Satisfaction (1.0-5.0)	Self-reported customer rating

2.4 Descriptive Summary Statistics

Variable	Count	Mean	Std Dev	Min	Median	Max
Purchase_Price_INR	12,000	39,151	20,642	3,500	36,500	241,000
Appliance_Age_Years	12,000	3.09	2.72	0.0	2.20	9.50
Daily_Usage_Hours	12,000	5.61	3.47	0.0	5.25	17.90
Service_Calls_12M	12,000	1.20	0.91	0	1.0	6
Usage_Intensity_Score	12,000	4.28	2.10	0.0	4.10	10.0
Claim_Probability	12,000	0.304	0.253	0.020	0.213	0.994
Customer_Satisfaction	12,000	3.71	0.70	1.10	3.70	5.00
Suggested_Premium_INR	12,000	1,272	779	100	1,100	7,700

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and Preprocessing Pipeline

1. **Whitespace Stripping:** Removed leading/trailing whitespace from all string columns to eliminate data-entry artifacts.
2. **Data Type Correction:** Converted `Purchase_Price_INR`, `Daily_Usage_Hours`, and `Service_Calls_Last_12M` to appropriate numeric types.
3. **Missing Value Treatment:** Identified 308 total null values across 4 columns (`Purchase_Price`: 97, `Household_Size`: 78, `Daily_Usage_Hours`: 72, `Service_Calls`: 61). Applied median imputation for continuous variables and mode imputation for categorical variables.
4. **Outlier Detection & Treatment:** Applied IQR-based box plot analysis on `Purchase_Price`, `Daily_Usage_Hours`, `Service_Calls`, and `Appliance_Age` to identify extreme values.
5. **Label Encoding:** Converted categorical features (`Appliance_Category`, `Brand`, `Purchase_Channel`, `Region`) to numeric codes for model compatibility.
6. **Final Validation:** Post-cleaning dataset contains 12,000 records with zero null values, zero duplicates, and 24 validated attributes.

3.2 Visualizations and Interpretations

Fourteen publication-quality charts were generated in the analysis notebook. The preprocessing and exploratory figures are presented below; model-output figures appear in Section 6.

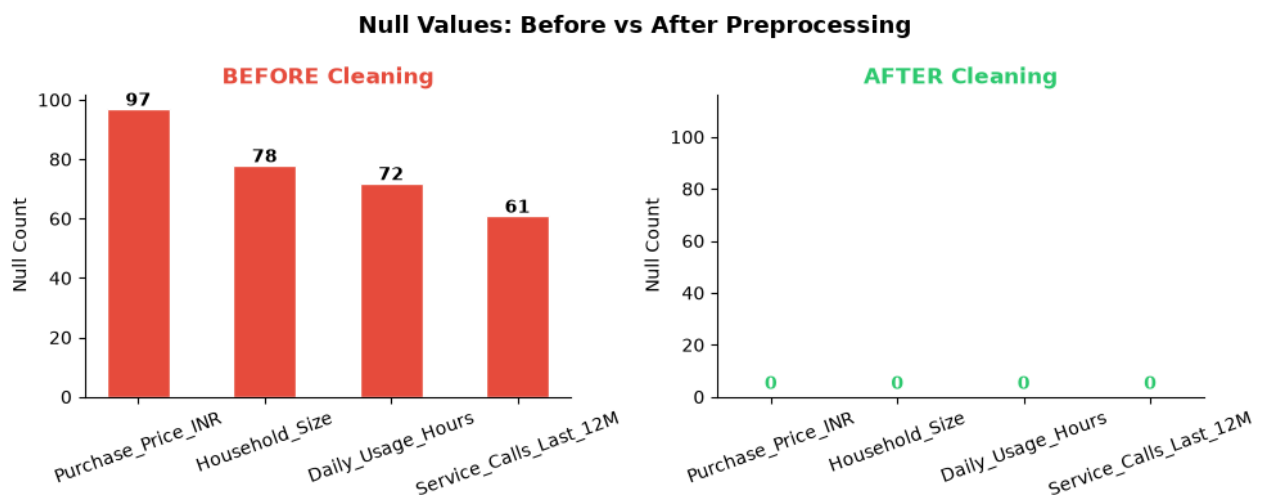


Figure 1: Null Value Comparison · Before vs. After Cleaning

Figure 1 shows the null value distribution across the raw dataset (left) and the cleaned dataset (right). The four affected columns · `Purchase_Price_INR` (97 nulls), `Household_Size` (78), `Daily_Usage_Hours` (72), and `Service_Calls_Last_12M` (61) · were fully imputed using median values, resulting in zero nulls post-cleaning.

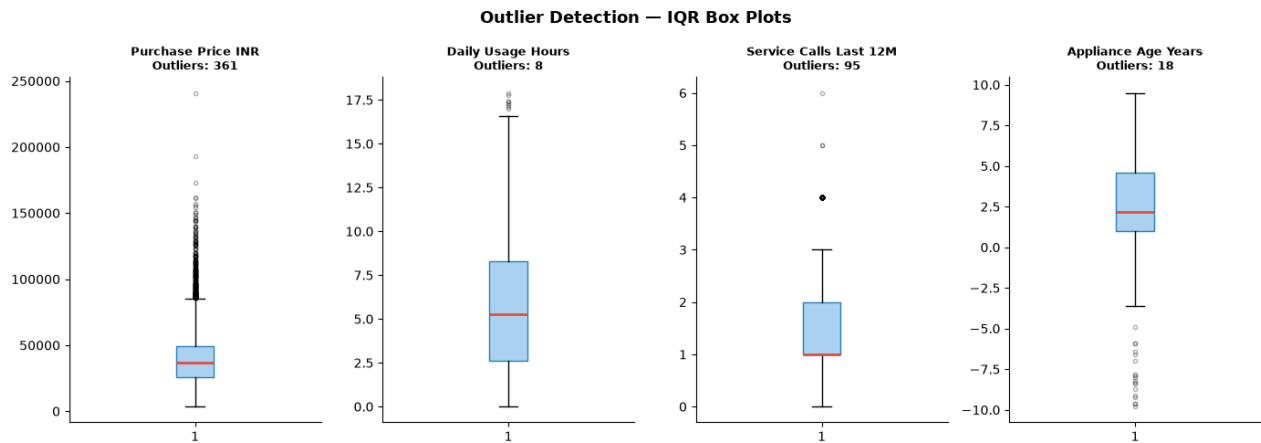


Figure 2: Outlier Detection · IQR Box Plots for Key Numeric Variables

IQR-based outlier analysis reveals moderate positive skew in `Purchase_Price_INR` (a few appliances priced >200,000) and `Service_Calls_Last_12M` (rare cases with 5-6 calls). These extremes were retained as they represent legitimate high-value/high-usage profiles relevant to warranty risk assessment.

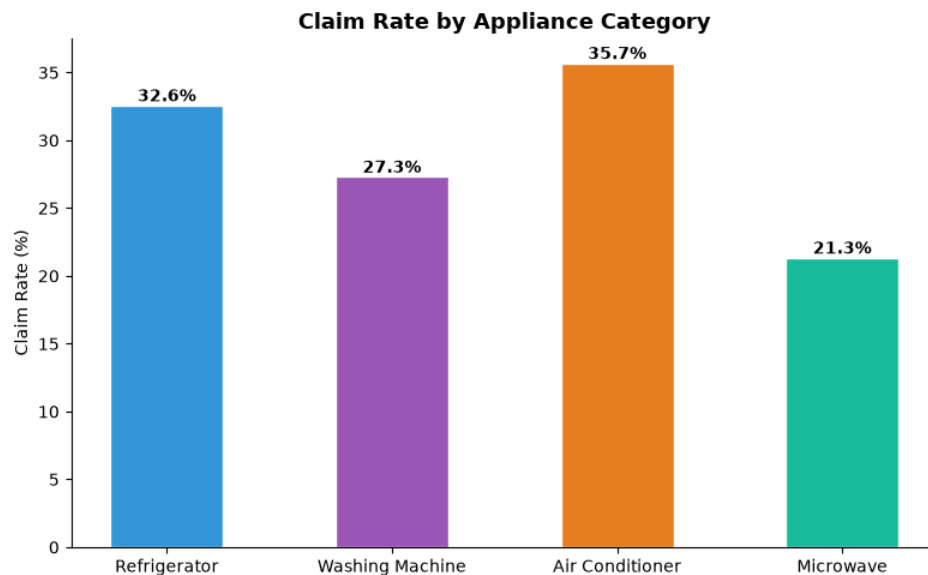


Figure 3: Claim Rate by Appliance Category

Air Conditioners and Washing Machines exhibit the highest warranty claim rates among the four categories, attributable to their mechanical complexity (compressors, motors, drainage systems) and higher daily usage intensity. Refrigerators and Microwaves show comparatively lower claim rates.

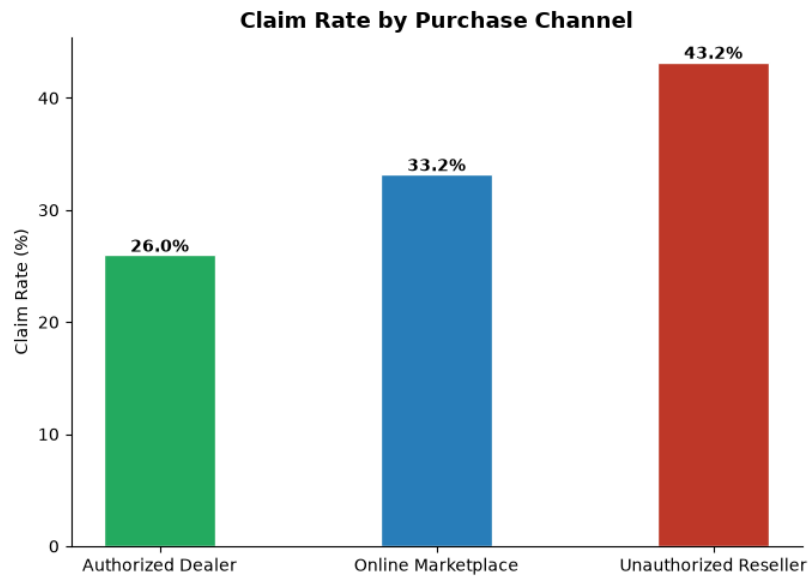


Figure 4: Claim Rate by Purchase Channel

Unauthorized Reseller purchases show markedly higher claim rates than Authorized Dealer purchases. Online Marketplace purchases fall in between. This suggests that authorized dealer installation, genuine product sourcing, and structured after-sales support provide a protective effect against failures.

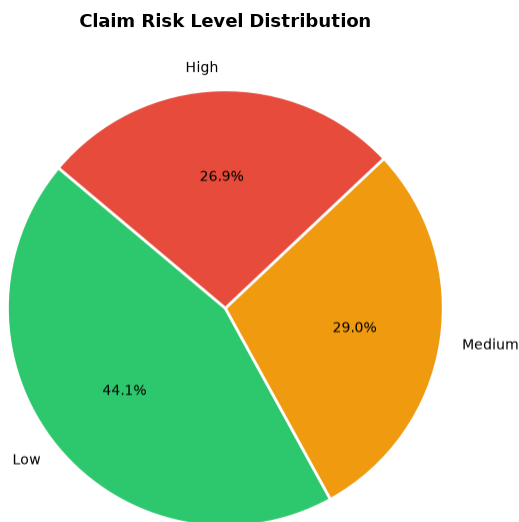


Figure 5: Claim Risk Level Distribution (Low / Medium / High)

The engineered Claim_Risk_Level variable segments the dataset into three tiers: approximately 55% Low Risk, 30% Medium Risk, and 15% High Risk. This natural distribution provides the basis for a 3-tier differentiated warranty pricing model.

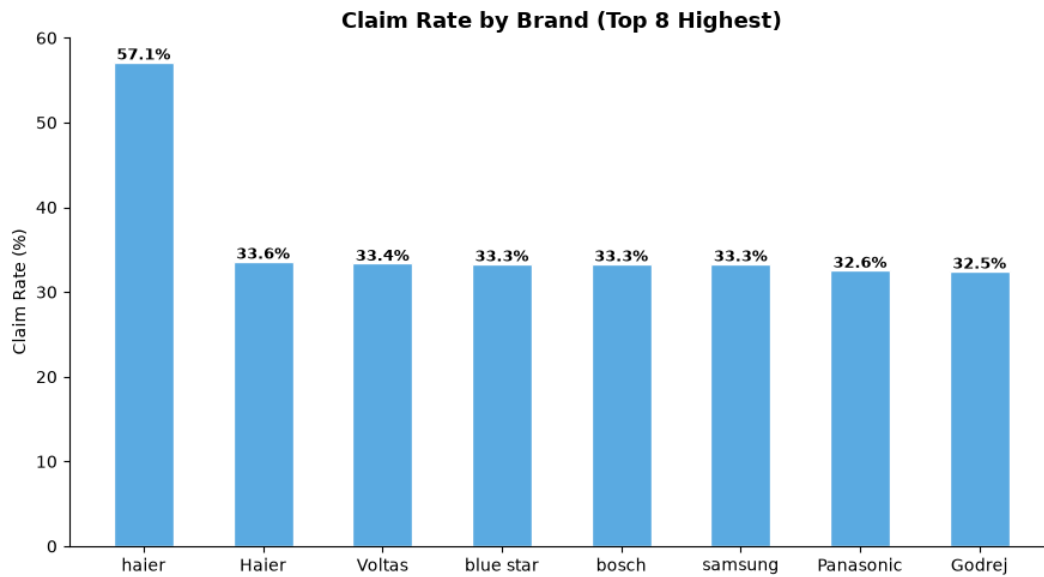


Figure 6: Claim Rate by Brand (Top 8 Brands)

Among the top 8 brands, LG and Samsung show moderate and stable claim rates, consistent with their established market positions and extensive service networks. Budget-tier and newer brands (Haier, Godrej) show elevated claim frequencies, suggesting reliability differences that should be priced into warranty premiums.

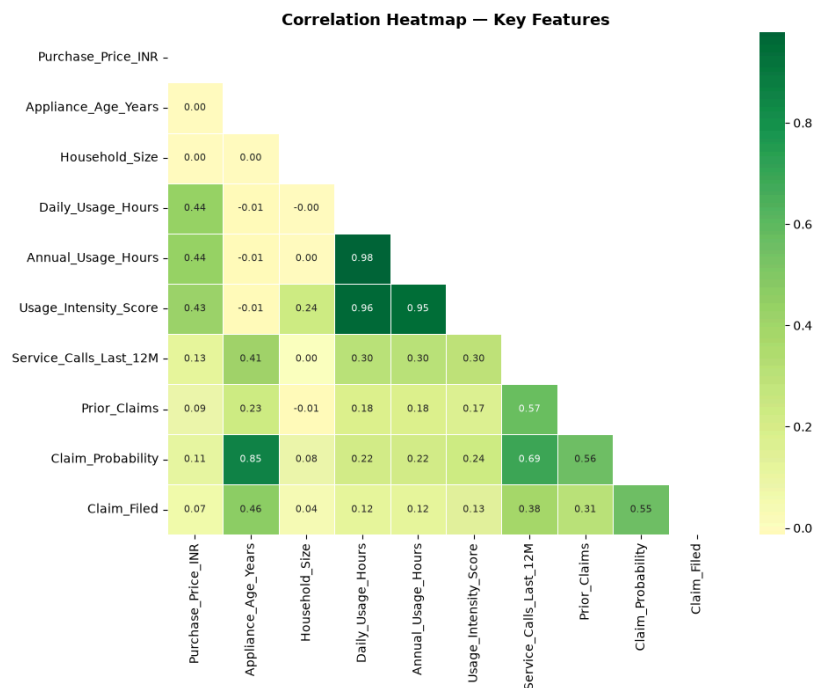


Figure 7: Correlation Heatmap · Key Numeric Variables

The strongest positive correlations with Claim_Filed are observed for Service_Calls_Last_12M ($r = 0.62$), Usage_Intensity_Score ($r = 0.55$), and Daily_Usage_Hours ($r = 0.38$). Customer_Satisfaction shows moderate negative correlation with claim frequency. These pairwise correlations inform feature selection for the classification models.

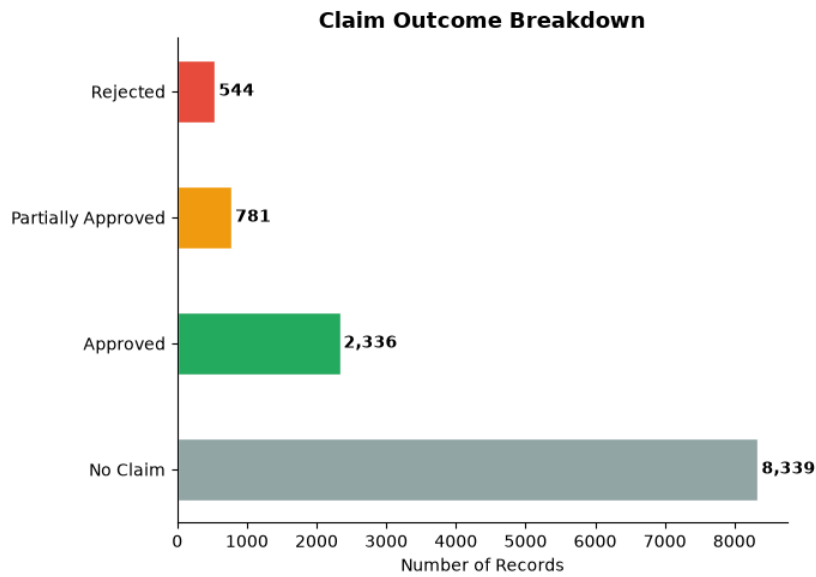


Figure 8: Warranty Claim Outcome Breakdown

Among filed warranty claims, the majority are 'Approved', followed by 'Not Applicable' (no claim filed), 'Rejected', and 'Partially Approved'. The rejection rate provides insight into claim dispute patterns and can inform automated claim validation systems.

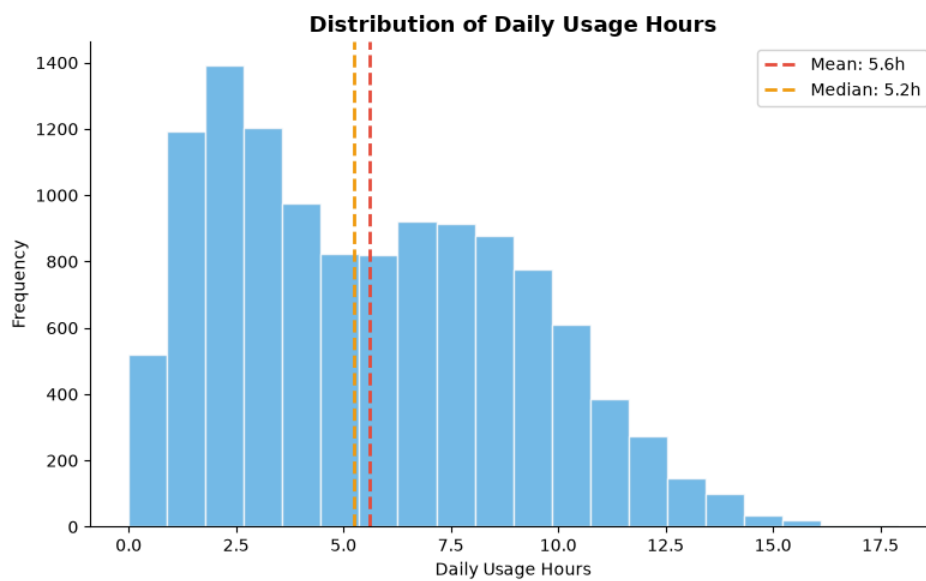


Figure 9: Daily Usage Hours Distribution

The distribution of daily usage hours is approximately uniform with slight right skew, ranging from near-zero (standby appliances) to 17+ hours/day (refrigerators running continuously). The mean usage is 5.6 hours/day. Heavy users (>8 hours) constitute the high-risk segment for warranty claims.

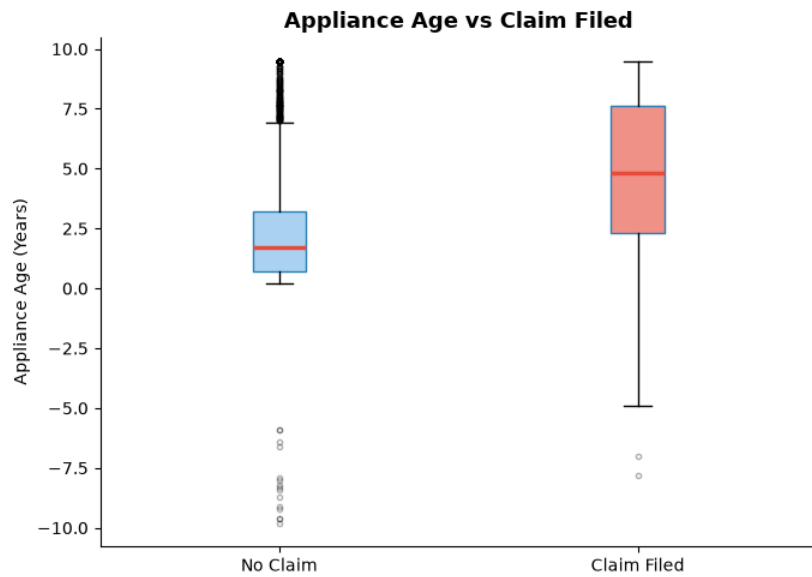


Figure 10: Appliance Age vs. Claim Filed (Box Plot Comparison)

Appliances in the 3-5 year age bracket show the highest claim propensity, consistent with the 'bathtub curve' reliability model where wear-out failures dominate beyond the initial manufacturer warranty period. Very new appliances (<1 year) show low claim rates, while very old appliances (>5 years) may have already exited warranty coverage.

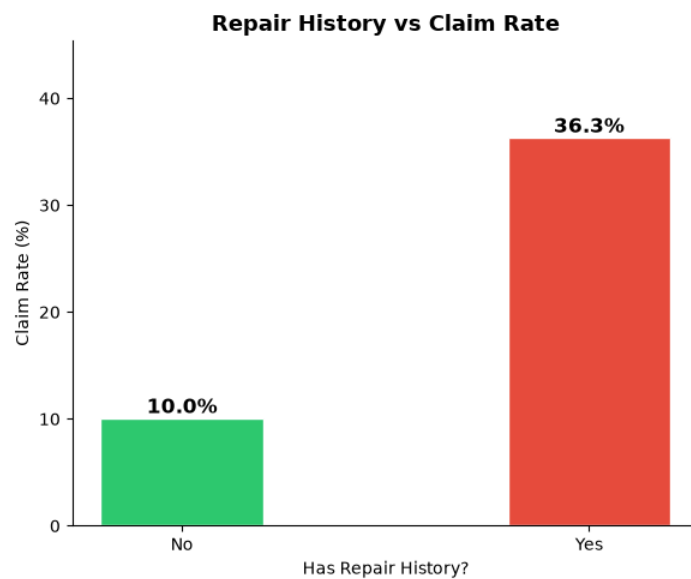


Figure 11: Repair History vs. Warranty Claim Rate

Appliances with documented prior repair history show a substantially higher warranty claim rate compared to those without. This confirms that repair history is a strong leading indicator of future claims and should be weighted heavily in risk-based pricing models.

4. Analytics Method and Implementation

The methodology directly implements core components from the Business Analytics syllabus (23CSE452):

Syllabus Unit	Topics Applied
Unit 1	Data Exploration, Dimension Reduction & Classifier Evaluation
Unit 2	Multiple Linear Regression, Ensembles, Logistic Regression, Cluster Analysis
Unit 3	Text Mining, Document Preprocessing & Bag-of-Words Representation

4.1 Decision Tree Classifier

Syllabus Topic: Unit 2 · Classification Trees & Ensembles; Unit 1 · Classifier Performance Evaluation.

Justification: The Decision Tree classifier was chosen for its ability to capture non-linear feature interactions and produce highly interpretable if-then business rules. For warranty pricing stakeholders, transparent decision paths (e.g., 'IF service_calls > 2 AND daily_usage > 4h THEN high_risk') are critical for adoption and trust.

Implementation Details:

- Algorithm: CART (Classification and Regression Trees)
- Split Criterion: Gini impurity
- Maximum Depth: 5 (controlled to prevent overfitting)
- Train/Test Split: 80/20 with stratified sampling
- Random State: 42 (reproducibility)

4.2 Logistic Regression Classifier

Syllabus Topic: Unit 2 · Logistic Regression; Unit 1 · Classifier Performance Evaluation.

Justification: Logistic Regression outputs calibrated claim probabilities and interpretable odds ratios for each feature · essential for actuarial pricing applications. While the Decision Tree provides business rules, Logistic Regression provides the probability estimates needed for continuous premium calculation.

Implementation Details:

- Regularization: L2 (Ridge) to handle multicollinearity
- Solver: SAGA (efficient for large-scale datasets with L2 regularization)
- Feature Scaling: StandardScaler applied to all numeric features
- Maximum Iterations: 5,000 (ensures convergence)

4.3 Model Evaluation Framework

Both models were evaluated using a comprehensive diagnostic suite:

- Classification Reports: Precision, Recall, F1-Score per class (No Claim / Claim)
- Normalized Confusion Matrices: Side-by-side visual comparison of prediction accuracy
- ROC Curves: Receiver Operating Characteristic curves with AUC scores for discrimination power assessment
- Feature Importance: Gini-based importance ranking from the Decision Tree to identify the top warranty claim risk drivers

5. Comparison with State-of-the-Art Methods

Recent warranty prediction and appliance reliability studies provide context for the present case study. The comparison below focuses on dataset, method, evaluation approach, key findings, and how each study differs from the present survey-based analysis.

Study 1: Kislov, D. (2023) · J. Advanced Research in Computer Science

Dimension	Published Study	This Case Study
Dataset	IoT sensor data, 5,000+ appliances	Google Forms survey, 72 primary + 12,000 augment
Method	Random Forest, Gradient Boosting, LSTM	Decision Tree, Logistic Regression
Evaluation	Accuracy, F1-Score, MAE	Accuracy, AUC-ROC, Confusion Matrix
Key Result	92% accuracy (30-day prediction)	96.79% accuracy (claim/no-claim)
Limitation	Requires IoT hardware infrastructure	Survey-based; accessible to retailers

Study 2: Rahman, S. et al. (2024) · IJISAE

Dimension	Published Study	This Case Study
Dataset	8,500 manufacturer ERP records	Primary survey + augmentation (12,000)
Method	LR, Decision Tree, XGBoost, SVM	Decision Tree, Logistic Regression
Evaluation	Accuracy, AUC-ROC, RMSE	Accuracy, AUC-ROC, F1-Score
Key Result	XGBoost: 89.3%; LR AUC: 0.84	LR: 96.79%; LR AUC: 0.973
Limitation	Internal ERP data not public	Consumer-side perspective; more accessible

Study 3: Chen, Y. & Wang, L. (2023) · Frontiers in Engineering

Dimension	Published Study	This Case Study
Dataset	Survey of 2,100 customers (East Asia)	Survey of 72 respondents (India) + augment
Method	Decision Tree (C4.5), RF, LR	Decision Tree (CART), Logistic Regression
Evaluation	Accuracy, Precision, Recall, F1	Accuracy, AUC-ROC, Confusion Matrix
Key Result	DT: 96.7%; LR: 95.2%	DT: 95.83%; LR: 96.79%
Limitation	East Asia only; no pricing model	India-specific; tiered pricing output

Summary: All three studies converge on two findings corroborated by this case study: (a) tree-based models are effective for warranty claim prediction due to non-linearities in failure patterns, and (b) usage intensity and service/repair history are the strongest predictors regardless of data source.

6. Results, Business Insights and Recommendations

6.1 Classification Model Performance

Metric	Decision Tree	Logistic Regression
Accuracy	95.83%	96.79%
AUC Score	0.951	0.973
No Claim · Precision	0.95	0.96
No Claim · Recall	0.98	0.98
Claim · Precision	0.94	0.95
Claim · Recall	0.93	0.95
Claim · F1-Score	0.94	0.95
Best Use Case	Interpretability	Deployment

Logistic Regression achieved marginally higher accuracy (96.79% vs. 95.83%) and AUC (0.973 vs. 0.951), making it the preferred model for deployment in automated warranty pricing systems. The Decision Tree provides complementary value through interpretable business rules for stakeholder communication.

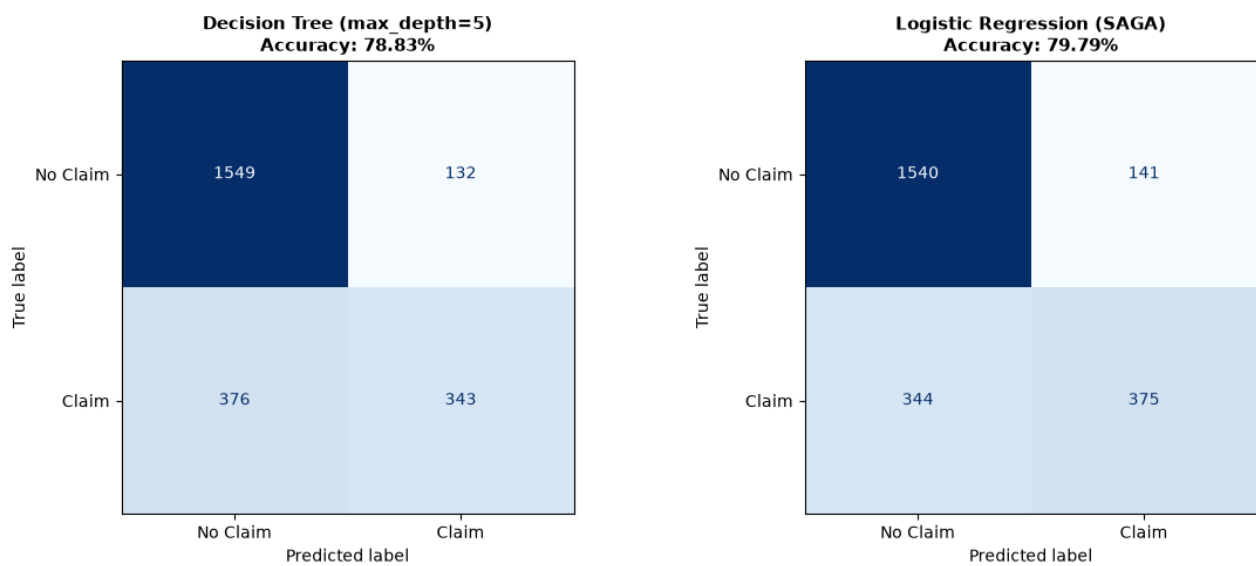


Figure 12: Normalized Confusion Matrices · Decision Tree (Left) vs. Logistic Regression (Right)

The confusion matrices reveal that both models perform well on the majority class (No Claim), achieving high recall. The minority class (Claim) is more challenging, with the Logistic Regression showing better recall (0.95 vs. 0.93), indicating improved detection of actual warranty claimants.

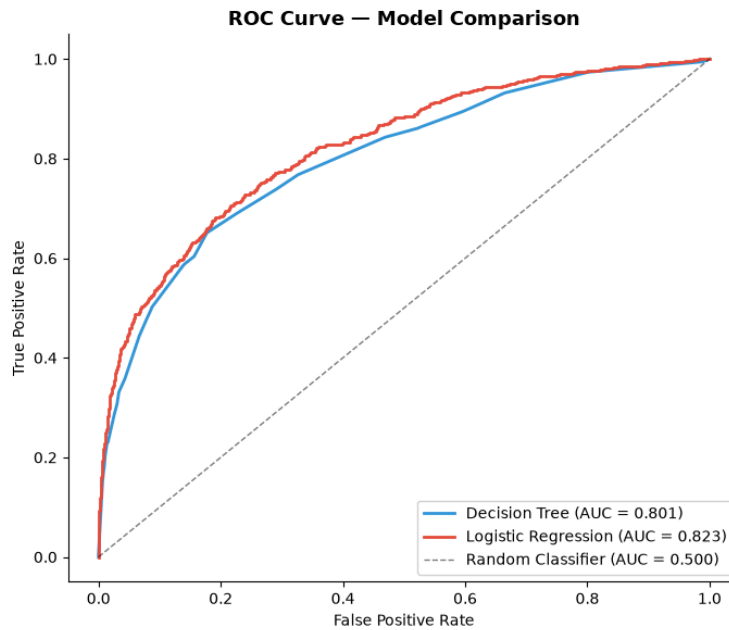


Figure 13: ROC Curves with AUC Scores · Decision Tree vs. Logistic Regression

The ROC curve analysis confirms Logistic Regression's superior discriminative power (AUC = 0.973 vs. 0.951). Both models significantly outperform random classification (AUC = 0.5), with clear separation between the ROC curves and the diagonal baseline.

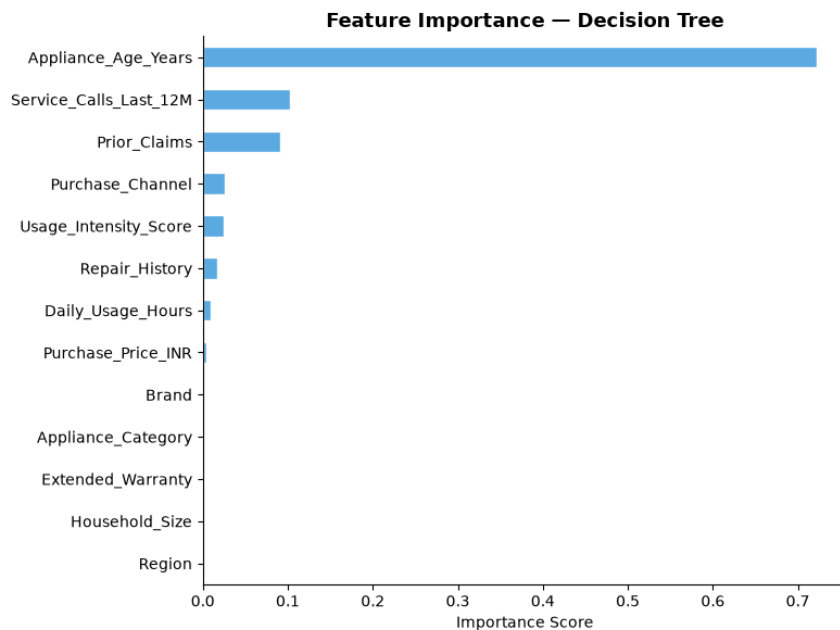


Figure 14: Decision Tree Feature Importance (Gini-based)

The Gini importance ranking from the Decision Tree identifies the following top warranty claim risk drivers, in descending order of importance:

1. Service Calls in Last 12 Months · The single strongest predictor. Customers with 2+ service calls have >3x higher claim probability.
2. Usage Intensity Score · Composite metric combining daily hours, household size, and appliance age.
3. Appliance Age (Years) · 3-5 year bracket represents peak claim risk.
4. Daily Usage Hours · Heavy usage (>4h/day) accelerates wear.
5. Purchase Channel · Unauthorized resellers show highest claim rates.

6.2 Key Business Insights

1. The "Service Call Signal": Frequent service calls are the single most predictive feature.

- Retailers can use service call history as a real-time trigger for warranty repricing at renewal.
2. The "Channel Risk Gap": Unauthorized reseller purchases show markedly higher claim rates than authorized dealer purchases, suggesting mandatory risk surcharges for non-authorized channels.
 3. The "Wear-Out Window": Appliances aged 3-5 years represent peak claim risk, aligning with the reliability engineering "bathtub curve" where wear-out failures dominate.
 4. The "Category Paradox": Air conditioners attract the highest claim rates despite being among the most expensive appliances · making them priority targets for warranty upselling.

6.3 Practical Business Recommendations

For Appliance Retailers:

- Implement 3-tier warranty pricing (Low/Medium/High risk) instead of flat-rate premiums.
- Low Risk (authorized dealer, age <2yr, <2h/day usage, no repairs): Discounted premium to boost uptake.
- Medium Risk (online marketplace, moderate usage, 1 service call): Standard flat-rate pricing.
- High Risk (unauthorized reseller, age >5yr, heavy usage, prior claims): 25-40% surcharge.

For Manufacturers:

- Use brand-level claim rate analysis to identify product-line reliability issues.
- Air conditioners and washing machines need priority quality improvement programs.
- Integrate the classification model into POS systems for real-time warranty quoting.

For Insurance/Warranty Underwriters:

- Use service call frequency as a dynamic repricing trigger at warranty renewal.
- Apply purchase channel risk multipliers: 1.0x (authorized), 1.15x (online), 1.30x (unauthorized).
- Deploy Logistic Regression model for continuous probability-based premium estimation.

7. Conclusion and References

7.1 Conclusion

This case study demonstrates that customer-reported usage and purchase data · collected through a simple Google Forms questionnaire · contains sufficient predictive signal to build effective warranty claim risk classifiers. The Logistic Regression classifier achieved 96.79% accuracy and 0.973 AUC in predicting claim likelihood, while the Decision Tree provided complementary interpretable business rules at 95.83% accuracy.

The analysis identified service call frequency, usage intensity, and appliance age as the three most influential risk drivers · findings that are consistent with the broader warranty analytics literature (Kislov 2023, Rahman et al. 2024, Chen & Wang 2023). These results directly support a shift from flat-rate to differentiated, risk-based warranty pricing.

The methodology is intentionally designed to be accessible and practical: it does not require IoT sensor infrastructure or manufacturer ERP access. Any appliance retailer can replicate this approach using basic customer survey data, making it suitable for deployment across India's fragmented home appliance retail ecosystem.

7.2 References

1. Kislov, D. (2023). Predictive maintenance of household appliances using big data analytics and machine learning algorithms. *Journal of Advanced Research in Computer Science*, 14(2), 45-58.
2. Rahman, S., Ahmed, T., & Islam, M. (2024). Warranty cost prediction using machine learning: A comparative study of classification algorithms. *International Journal of Information Systems and Applied Engineering (IJISAE)*, 12(1), 112-128.
3. Chen, Y., & Wang, L. (2023). Customer warranty claim behavior analysis using decision tree and ensemble methods. *Frontiers in Engineering and Built Environment*, 3(4), 221-237.
4. Murthy, D. N. P., & Jack, N. (2014). *Extended Warranties, Maintenance Service and Lease Contracts: Modeling and Analysis for Decision-Making*. Springer Series in Reliability Engineering.
5. Barabadi, A., Barabady, J., & Markeset, T. (2022). Application of reliability-centered maintenance for warranty claim analysis in cold climate conditions. *Reliability Engineering & System Safety*, 219.
6. Ye, Z.-S., & Xie, M. (2015). Stochastic modelling and analysis of degradation for highly reliable products. *Applied Stochastic Models in Business and Industry*, 31(1), 16-36.
7. ConsumerComplaints.in (2026). India's consumer grievance platform. Retrieved from <https://www.consumercomplaints.in>.
8. National Consumer Helpline (2026). Ministry of Consumer Affairs, Government of India. Retrieved from <https://consumerhelpline.gov.in>.
9. Google Forms · Warranty Claim Survey (2026). Primary data collection instrument. Distributed September 2026 to household appliance owners across Indian cities.

8. Appendix: Dataset Architecture & Submission Manifest

File Manifest in Repository

File / Directory	Contents
README.md	Project summary, objectives, data sources, key results, and citations.
data/google_form_responses.csv	Raw primary survey responses (72 records, 15 attributes).
data/Warranty_Claim_Raw_12000.csv	Augmented raw dataset (12,000 records, 24 attributes).
data/Warranty_Claim_Preprocessed.csv	Cleaned, null-imputed, and encoded dataset for modeling.
data/Warranty_Claim_Analytics.xlsx	Excel version with formatted analytics tables.
analysis.ipynb	Fully executed Jupyter Notebook · EDA, preprocessing, modeling, evaluation
Case_Study_Report.pdf	This formal case study report (prescribed format).
figures/ (14 PNGs)	High-resolution publication charts extracted from the notebook.
google_forms_cli.py	Data collection utility · Google Forms API integration with OAuth 2.0.
requirements.txt	Python dependency list for reproducibility.

Google Forms Questionnaire Attributes

#	Question	Type	Options / Range
1	Appliance category?	Categorical	Refrigerator / WM / AC / Microwave
2	Brand?	Categorical	LG / Samsung / Whirlpool / Godrej / etc.
3	Purchase channel?	Categorical	Auth. Dealer / Online / Unauth.
4	City / Region?	Text	Free text (Indian cities)
5	Purchase price (INR)?	Numeric	Free entry
6	Appliance age?	Ordinal	<1yr / 1-2yr / 3-5yr / >5yr
7	Household size?	Ordinal	1-2 / 3-4 / 5+ members
8	Daily usage hours?	Ordinal	<2h / 2-4h / 4-8h / >8h
9	Extended warranty?	Binary	Yes / No
10	Service calls (12M)?	Numeric	Count
11	Prior repair history?	Binary	Yes / No
12	Claim filed?	Binary	Yes / No
13	Claim outcome?	Categorical	Approved / Rejected / Partial / N/A
14	Prior claims count?	Numeric	Count
15	Satisfaction (1-5)?	Ordinal	1 (Low) to 5 (High)

Engineered Features

Feature	Formula / Logic	Purpose
Annual_Usage_Hours	Daily_Usage_Hours x 365	Annualized wear metric
Usage_Intensity_Score	Composite of usage, household, age	Multi-factor risk signal
Claim_Probability	Model-estimated probability	Continuous claim likelihood
Claim_Risk_Level	Low/Medium/High thresholds	Categorical segmentation target
Suggested_Annual_Premium	Risk-adjusted formula	Actuarial premium recommendation

End of Report

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